



Acute Coronary Syndromes

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Introduction

The term acute coronary syndrome (ACS) refers to any clinical symptoms compatible with acute myocardial ischemia. These include unstable angina and non-ST segment elevation myocardial infarction (NSTEMI), which are also referred to collectively as non-ST elevation acute coronary syndromes (NSTEMACS) and ST segment elevation myocardial infarction (STEMI). Treatment centres should have a standardized approach to patients with ACS to ensure the most rapid assessment and initiation of treatment possible.

The following discussion relates to any ACS that is associated with primary coronary events caused by plaque erosion and/or rupture, fissuring or dissection.

Unstable Angina and Non-ST Segment Elevation Myocardial Infarction

Goals of Therapy

- Decrease mortality and complications
- Reduce the severity or eliminate episodes of ischemia
- Prevent further myocardial injury

Investigations

- Careful history with special attention to pain (quality, severity, location, radiation, precipitating and relieving factors), duration of symptoms, previous cardiac history, cardiac risk factors (smoking status, diabetes mellitus, hyperlipidemia, hypertension, family history of first-degree relative with MI before 55 years of age if male or before 65 if female) and effect of nitroglycerin
- Physical examination with attention to the presence of hypertension, heart failure or valvular heart disease
- Laboratory tests:
 - ECG, CBC, electrolytes, glucose, creatinine and lipid profile (within 24 hours of presentation)
 - troponin is a highly accurate, sensitive and specific indicator of myocardial injury.^[1] Measurement allows reliable stratification of risk and prediction of outcomes in individual patients.^{[2][3]} Elevation of cardiac troponin levels may also result from pulmonary embolus or nonischemic mechanisms of myocardial injury, such as myocarditis, severe heart failure or cardiac trauma. Additionally, elevated troponins are present in 30–50% of patients with pericarditis as a result of epicardial inflammation, and have also been reported in critical illness, sepsis, neurologic events, hypothyroidism, chemotherapy-induced myocardial toxicity and renal insufficiency. Note that biotin supplements may interfere with troponin testing, leading to falsely low values; consult the testing laboratory to confirm whether biotin can alter the patient's troponin test result.^[4]
- Careful search for secondary causes of ischemia, e.g., anemia, fever, infection, arrhythmia, thyroid disease
- Echocardiography can be used early when clinical history and ECG are nondiagnostic—the presence of regional wall motion abnormalities with chest pain is suggestive of underlying ischemia

The TIMI (Thrombolysis in Myocardial Infarction) risk score is a risk stratification and decision support tool for patients with unstable angina (UA) or NSTEMI (UA/NSTEMI) using clinical features present at the time of initial assessment in the emergency department (see [Table 1](#)). It predicts the risk of both death and early recurrent ischemic events and is used to tailor different evidence-based therapies to appropriate patients. As the risk score increases, so too do adverse outcomes.^[5]

Table 1: Clinical Features Used to Calculate TIMI Risk Score^[5]

1 point is assigned for each of the following clinical features:^[a]

- >65 y of age
- ≥3 cardiac risk factors (hypercholesterolemia, hypertension, diabetes mellitus, current smoker, family history of coronary artery disease)
- ≥50% coronary artery stenosis
- Any ASA use within the past 7 days
- ≥2 episodes of angina within the past 24 h

- Elevation in cardiac markers (troponin or creatine kinase-myocardial band)
- ST segment deviation ≥ 0.5 mm on ECG

^[a] The predicted risk of an adverse cardiovascular event (all-cause mortality, new or recurrent myocardial infarction, ischemia requiring urgent revascularization) 14 days later is as follows: 0/1 point = 4.7%, 2 points = 8.3%, 3 points = 13.2%, 4 points = 19.9%, 5 points = 26.2%, 6/7 points = 40.9%

The [Global Registry of Acute Coronary Events \(GRACE\) score](#) is a more comprehensive decision support tool. While more complicated, a web-based tool simplifies risk calculation. When compared to the TIMI risk score, this model has been found to better discriminate risk of 1-month and 1-year mortality in patients with ACS.^[6]

Therapeutic Choices

The management of patients with UA/NSTEMI has evolved significantly (see [Figure 1](#)). Evidence from randomized clinical trials strongly supports the use of the “invasive strategy”: early, urgent coronary angiography followed, if possible, by revascularization with percutaneous coronary intervention (PCI) or bypass surgery in all high-risk patients. Patients are at high risk if they have 1 or more of the following features: positive cardiac enzymes, ST segment changes, TIMI risk score ≥ 3 , recurrent ischemic symptoms, heart failure, hemodynamic instability, sustained ventricular tachycardia, or a prior revascularization procedure such as a coronary artery bypass graft (CABG) or PCI.^[7]

Nonpharmacologic Choices

All patients admitted with UA/NSTEMI should be placed on bed rest while ischemia is ongoing, then gradually mobilized when symptoms have stabilized. Continuous ECG monitoring for potentially lethal arrhythmias and ST segment shifts (if available) is indicated in all high-risk patients.

Pharmacologic Choices

[Table 2](#) provides details on the medications used in UA/NSTEMI.

Nitrates

Initial attempts at symptom relief should involve the use of **nitroglycerin** sublingual tablets or spray. IV nitroglycerin is indicated in patients whose symptoms are not relieved promptly (within 15–20 minutes). Longer-acting oral or topical nitrates can be used when patients are symptom-free to prevent recurrent episodes of ischemia. The use of sildenafil or vardenafil in the previous 24 hours, tadalafil in the previous 48 hours or the presence of significant hypotension are contraindications to the use of nitrates.

Beta-adrenergic Antagonists (Beta-blockers) and Calcium Channel Blockers

Start **beta-blockers** as soon as possible in all patients without contraindications: reactive airway disease, bradycardia (heart rate ≤ 50 bpm), second- or third-degree heart block without a functioning pacemaker, hypotension (systolic blood pressure [SBP] < 100 mm Hg). The dose should be titrated as tolerated to achieve target doses. If anginal pain is ongoing at presentation, beta-blockers can be initially administered intravenously, followed by oral therapy.

Calcium channel blockers can be used to control ongoing symptoms of ischemia in patients who are receiving maximum tolerated doses of beta-blockers and adequate doses of nitrates. In addition, these agents are used in patients who cannot tolerate beta-blockers and in those with variant angina (coronary spasm). Nondihydropyridine calcium channel blockers (e.g., **diltiazem**, **verapamil**) should be used with caution when administered with beta-blockers to avoid left ventricular (LV) dysfunction, severe bradycardia or increasing atrioventricular (AV) nodal block. Avoid immediate-release nifedipine because the acute drop in blood pressure increases risk of stroke.^[8]

Angiotensin-Converting Enzyme Inhibitors

ACE inhibitors reduce mortality in patients with LV systolic dysfunction or clinical heart failure, diabetes, or recent MI, and in a broad spectrum of patients with high-risk chronic coronary artery disease.^{[9][10]} Therefore, use these agents (ideally initiated within 24 hours of presentation) in all of these patient groups and in patients with hypertension not controlled with beta-blockers and nitrates.

Anticoagulation Therapy

Heparins and Fondaparinux

Heparin is a key component in the successful management of patients with ACS. Studies of ASA with either unfractionated heparin (UFH) or low-molecular-weight heparins (LMWH) have shown a 3–3.5% absolute reduction (50–60% relative reduction) in the rate of death or MI in the first week.^{[11][12]} UFH has important limitations due to significant variability in anticoagulant response. LMWHs have the advantage of ease of administration, predictable anticoagulant response and no need for monitoring of anticoagulant effect. Clinical trials have demonstrated the superiority of **enoxaparin** over UFH, but have shown neutral or unfavourable trends with other LMWHs (**dalteparin** and **nadroparin**).^{[13][14][15][16]} A direct comparison favoured enoxaparin over **tinzaparin**.^[17] Enoxaparin is therefore the preferred agent in patients with UA/NSTEMI who do not have significant renal dysfunction (estimated CrCl > 30 mL/min). In those with an estimated CrCl ≤ 30 mL/min, UFH is appropriate. The optimal duration of heparin therapy is unknown but generally is 2–5 days. Heparin is usually not administered following successful PCI.

Fondaparinux, an indirect inhibitor of factor Xa, is as effective as enoxaparin in patients with NSTEMI ACS, so it can be considered as an alternative anticoagulant in this setting. It is associated with a lower incidence of major bleeding than enoxaparin according to the results of a large randomized trial.^[18]

Antiplatelet Therapy

ASA

ASA provides a significant benefit in patients with ACS.^[19] If patients with suspected UA/NSTEMI are not already receiving ASA, it should be initiated promptly and continued long term. ASA is routinely combined with a second antiplatelet from a different class (see [Figure 1](#)).

Thienopyridines

Thienopyridine antiplatelet agents block the P2Y₁₂ platelet receptor. In the CURE (Clopidogrel in Unstable angina to prevent Recurrent ischemic Events) trial, **clopidogrel** (plus **ASA**) significantly reduced the incidence of major adverse cardiac events in patients with ACS compared with ASA alone, and reduced the incidence of recurrent ischemia, an effect which was evident within a few hours.^[20] The benefit of clopidogrel was consistent among patients, regardless of their TIMI risk score.^[21] The major benefits were noted at 30 days, with small additional benefits observed over the subsequent treatment period (average 8 months). There was an excess of both major and minor bleeding in the clopidogrel group during the trial, with a nonsignificant trend towards an increase in life-threatening bleeding. Bleeding risks increase with higher ASA doses.^[22] Clopidogrel should be held for a minimum of 5 days in patients scheduled for non-urgent bypass surgery.^{[20][21]}

Regulatory authorities have noted the diminished effectiveness of clopidogrel in patients who are unable to convert it to its active form. Patients with diminished CYP2C19 function due to genetic polymorphisms activate clopidogrel poorly and have higher cardiovascular event rates, both following an ACS and after PCI. Tests are available to identify these individuals. However, there is little evidence that increasing the dose of clopidogrel or changing to another agent improves outcomes overall. For this reason, routine genetic testing is not recommended at this time.^[23]

Prasugrel is a more potent platelet inhibitor than clopidogrel and, in a large clinical trial of ACS patients undergoing PCI, was associated with lower ischemic event rates including stent thrombosis. However, there was a significant increase in bleeding events, particularly in those older than 75 years of age, those with body weight <60 kg, and those with a history of stroke or transient ischemic attack.^[24] The use of prasugrel should be considered in those patients with ACS undergoing stent implantation who are at higher risk for stent thrombosis (previous stent thrombosis, STEMI, history of diabetes mellitus). The TRILOGY-ACS study did not demonstrate superiority of prasugrel over clopidogrel in ACS patients who are managed medically.^[25] Interestingly, a randomized trial comparing prasugrel and ticagrelor in ACS patients planned for invasive evaluation demonstrated a lower risk of death, myocardial infarction and stroke in patients treated with prasugrel with no difference in bleeding.^[26] Prasugrel should not be used unless coronary anatomy has been defined; however, in cases where it has been started and CABG is considered optimal therapy, it should be held for a minimum of 7 days prior to surgery.

Cyclopentyltriazolopyrimidines

Cyclopentyltriazolopyrimidines also block the P2Y₁₂ receptor. **Ticagrelor** is more potent and has a faster speed of onset than clopidogrel, but does not require metabolic activation. In a large clinical trial of patients with ACS, the use of ticagrelor was associated with both lower mortality and ischemic events compared to clopidogrel, but with an increase in non-CABG-related bleeding.^[27] Ticagrelor is recommended as first-line therapy in patients with ACS.^{[19][27]} Ticagrelor should be held for a minimum of 5 days prior to non-urgent CABG. Dyspnea may be associated with ticagrelor therapy and may result in the need to switch to a thienopyridine.

Therapeutic Tips

- The standard dose of **ASA** is 81 mg daily.
- Discontinue **clopidogrel** or **ticagrelor** 5 days prior and **prasugrel** 7 days prior to non-urgent bypass surgery to decrease the risk of bleeding.

ST Segment Elevation Myocardial Infarction

Urgency

STEMI represents the extreme of the ACS spectrum and is considered a medical emergency. It therefore requires urgent assessment and treatment.

Goals of Therapy

- Decrease mortality and complications
- Reduce or contain infarct size
- Salvage functioning myocardium and prevent remodelling
- Re-establish patency of the infarct-related artery

Investigations

- Rapid, targeted history and physical examination with particular attention to onset of symptoms, contraindications to use of thrombolytic agents (see [Table 2](#)) and evidence of high-risk features (tachycardia, hypotension, heart failure)

- ECG STAT, then every 8 hours for the first 24 hours, then daily for 3 days. In addition, repeat the ECG with each recurrence of chest pain
- Baseline troponin STAT, then every 8 hours until enzymatic confirmation of the diagnosis (see UA and Non-ST Segment Elevation Myocardial Infarction, [Investigations](#) for further discussion of troponin)
- CBC to rule out the presence of anemia and establish baseline platelet count, baseline electrolytes, creatinine, fasting lipid profile (within 24 hours of presentation) and liver enzyme tests
- Portable chest x-ray (CXR) STAT
- Echocardiography to assess LV function after stabilization and treatment; echocardiography is also used emergently when there is suspicion of acute mechanical complications post-MI

Therapeutic Choices

The care of patients with STEMI is summarized in [Figure 2](#).

Nonpharmacologic Choices

Place all patients on bed rest with supplemental oxygen and continuous ECG monitoring. Begin gradual mobilization after stabilization, provided there is no evidence of complications. The use of routine supplemental oxygen in normoxic patients is not recommended as it may increase infarction size.^{[28][29]}

Primary Percutaneous Coronary Intervention

Urgent coronary angiography with PCI is considered first-line therapy for all patients who can access appropriate high-volume cardiac catheterization facilities.^[30] Primary PCI is indicated in patients with contraindications to thrombolytic therapy or cardiogenic shock, and is preferred in patients over 75 years of age because of a higher risk of intracranial hemorrhage and higher overall early mortality seen with thrombolytic agents in this age group.^{[31][32][33]} For all patients who cannot access PCI within 2 hours, a pharmacoinvasive strategy (involving administration of thrombolytic therapy followed by either immediate or more routine transfer to a cardiac catheterization centre) is now recommended as a first-line approach.^[30] Heparin agents are used post-PCI at the discretion of the treating physician.

Pharmacologic Choices

[Table 2](#) provides details on the medications used in STEMI.

Fibrinolytic Therapy

Fibrinolytic therapy (**alteplase** or **tenecteplase**) administered early in the course of STEMI substantially reduces morbidity and mortality, particularly if the patient presents within 6 hours of symptom onset.^[30] The benefit is reduced for patients who present 6–12 hours after the onset of symptoms, and evidence of

benefit is much less clear for those who present more than 12 hours after the onset of symptoms. The greatest absolute benefit is seen where the risk of mortality is highest (anterior MI, previous MI).

Heparins

Administer IV **unfractionated heparin (UFH)** to all STEMI patients receiving a fibrinolytic. **Enoxaparin** has been shown to be superior to UFH in patients treated with tenecteplase. It must be used with caution in patients older than 75 years of age (dosage adjustment required) or those with renal insufficiency (estimated CrCl <30 mL/min) because of an increased risk of bleeding.^[34] Heparins should be continued for a minimum of 48 hours and use can be extended in patients with high-risk features.

ASA

ASA significantly reduces mortality in patients with STEMI. If patients are not already receiving ASA, it should be initiated promptly and continued long term.^[30] If appropriate, consider adding a PPI in patients with a history of GI bleeding.

Beta₁-adrenergic Antagonists (Beta-blockers)

Beta-blockers are recommended to reduce the risk of recurrent MI in all STEMI patients without contraindications, and are particularly useful when sinus tachycardia and hypertension are present, or in the case of inferior STEMI, once bradycardia has resolved. They should be initiated once hemodynamic stability is achieved. Titrate doses to a resting heart rate of 50–60 bpm.

Calcium Channel Blockers

Calcium channel blockers increase morbidity and mortality in patients with STEMI and are not recommended.^[35] They may be used *cautiously* to relieve ischemia or to achieve rate control in patients with atrial fibrillation if beta-blockers are contraindicated. Consider low-dose **diltiazem**, e.g., 90–120 mg daily in divided doses, with heart rate monitored closely.

Nitrates

Unlike UA/NSTEMI, IV, oral or topical **nitrates** should be used in patients with STEMI only if ischemia is persistent or recurrent, or if the patient has a large anterior MI, hypertension or heart failure.^[36]

ACE Inhibitors and Angiotensin Receptor Blockers

ACE inhibitors are routinely recommended, unless contraindicated, in all patients post-STEMI and should ideally be started within 24 hours of the event unless the patient is hypotensive (SBP <100 mm Hg).^[30] Doses should be increased every 24 hours as tolerated for inpatients and at 1- to 2-week intervals for outpatients. Based on the results of the HOPE trial, it is reasonable to continue an ACE inhibitor indefinitely, even in patients with preserved LV function.^[10] An angiotensin receptor blocker (ARB) should

be used in STEMI patients who cannot tolerate an ACE inhibitor and have either clinical or radiological signs of heart failure or documented LV dysfunction.

Mineralocorticoid Receptor Antagonists

Patients receiving an ACE inhibitor and a beta-blocker may still have clinical evidence of heart failure, LV ejection fraction <40% or both. In these patients, consider adding a mineralocorticoid receptor antagonist (aldosterone antagonist) such as **spironolactone** or **eplerenone**, as they have been shown to reduce mortality.^[37] Due to the risk of hyperkalemia, use with caution in patients with renal insufficiency and in those receiving an ACE inhibitor or ARB. Suggestions for patient monitoring can be found in [Heart Failure](#).

Clopidogrel

In patients with STEMI treated with fibrinolysis, **clopidogrel** combined with **ASA** increases patency of the infarct-related artery and decreases ischemic complications according to the results of 2 randomized trials;^{[38][40]} therefore, clopidogrel should be administered at presentation. Ticagrelor has not been studied in those >75 years of age treated with fibrinolysis, but seems to be associated with similar rates of bleeding as clopidogrel.^[39]

Complications of STEMI

Recurrent or Ongoing Ischemia

Optimize antianginal therapy, and consider urgent coronary angiography and possible revascularization for all patients with recurrent or ongoing ischemia.

Heart Failure

Treat heart failure aggressively (see [Heart Failure](#)). In addition, consider angiography and possible revascularization in patients with LV ejection fraction <40% after MI.

Arrhythmias

Asymptomatic premature ventricular contractions (PVCs) do not require therapy. Symptomatic ectopy may require therapy, usually with a beta-blocker. Class IC agents are contraindicated. Patients with sustained ventricular arrhythmias require investigation to rule out recurrent ischemia and may require subsequent electrophysiologic assessment.^[41] Consider urgent electrical or pharmacologic cardioversion in patients with atrial fibrillation causing ischemia or heart failure. Treat all patients with atrial fibrillation to control ventricular rate (usually with a beta-blocker or digoxin if there is concomitant heart failure) and consider oral anticoagulant therapy.

Patients with an ejection fraction <30% 1 month post-MI or 3 months post-revascularization should be referred for automatic implantable cardioverter-defibrillator therapy (AICD) for prevention of sudden cardiac death.^[42]

Pericarditis

Pericarditis, although uncommon, usually presents within 72 hours post-MI, and symptoms usually resolve within 3–4 days. Symptomatic pericarditis can be treated by increasing the dose of **ASA** to 650 mg QID for 1–2 weeks. **Colchicine** can be added to ASA for patients with ongoing symptoms despite conservative measures. Discontinuation of anticoagulants is unnecessary if pericarditis occurs early postinfarction, but caution is required if a patient presents with pericarditis weeks or months after MI (Dressler syndrome) because of the risk of pericardial bleeding and tamponade.^[43]

Mechanical Complications

Ruptured papillary muscle (and severe mitral regurgitation), ventricular septal defects or ventricular free wall rupture are infrequent but devastating problems that can present with acute cardiac arrest or cardiogenic shock. Patients should be considered for emergency angiography and surgical intervention, but even with urgent treatment, the mortality rate associated with these conditions is very high.

Choices during Pregnancy and Breastfeeding

Acute Coronary Syndromes and Pregnancy

Acute coronary syndromes during pregnancy or early postpartum are rare but becoming more common as women delay childbearing and with an increase in risks such as diabetes and obesity. If a patient presents with suspected ACS during this time, referral for specialist assessment and management is recommended. Coronary artery disease will be the etiology in approximately 40% of cases. Consideration should be also be given to other etiologies such as coronary artery dissection.^{[44][45]} The following discussion pertains only to those patients where coronary artery disease with acute plaque rupture is present.

Management

A pregnant patient presenting with suspected ACS should be worked up in the same timely manner as a non-pregnant patient to avoid treatment delays that can contribute to adverse outcomes. Management should be collaborative and include obstetric and cardiovascular specialists. Discussions regarding the need to terminate pregnancy or the management of labour and delivery are outside the scope of this chapter.

Coronary angiography should be considered, as it will delineate the cause of the suspected ACS and therefore determine appropriate management. ASA at usual doses (81 mg/day) is safe in pregnancy. Clopidogrel is the preferred thienopyridine – other agents have not been studied in pregnancy. UFH and

LMWH can be used, recognizing the increased risk of maternal bleeding. Beta-blockers and nitrates agents can be used but given the risk of teratogenicity, ACE inhibitors, angiotensin receptor blockers and statin agents are contraindicated.

In the setting of STEMI, primary PCI is the preferred treatment, but recent evidence suggests that fibrinolysis may be an alternative where primary PCI is not available.^[46]

Acute Coronary Syndromes and Breastfeeding

The workup of breastfeeding women should again proceed in a timely fashion and not divert from usual care. ASA, UFH, LMWH and beta-blockers are considered safe for breastfeeding, but other cardiovascular medications used in the setting of ACS may not be. Therefore, in the interest of improving maternal outcomes, breastfeeding should be discontinued.

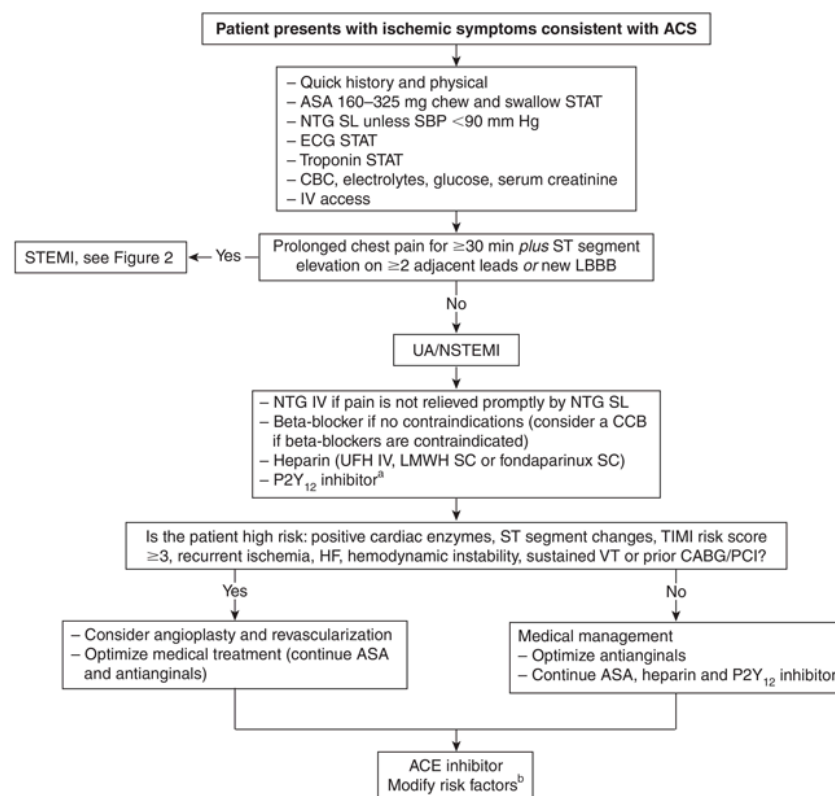
A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- The goal for thrombolytic treatment is a door-to-needle time of 30 minutes or less.
- The goal for primary PCI is a first-medical-contact-to-device time of 90 minutes or less.
- In patients with right ventricular infarcts:
 - avoid nitrates and diuretics
 - use fluids and inotropes to treat hypotension
- Administer beta-blockers in the first 24 hours to all patients without contraindications. Increase the dose every 12 hours (every 24 hours for once-daily beta-blockers), if tolerated (monitor blood pressure and heart rate), until the patient has reached adequate beta-blockade (HR $\leq$ 55–65 bpm).
- Start ACE inhibitors within 24 hours post-MI. The choice of agent can depend on practitioner preference, hospital formulary or financial constraints for the individual patient.
- In smokers, reinforce early (within 24 hours) and frequently the need to quit smoking.
- Stool softeners may be used in the immediate post-MI period to prevent straining with bowel movements.
- Anxiolytics may be used on an as-needed basis in the immediate post-MI period.

Algorithms

Figure 1: Early Management of UA/NSTEMI

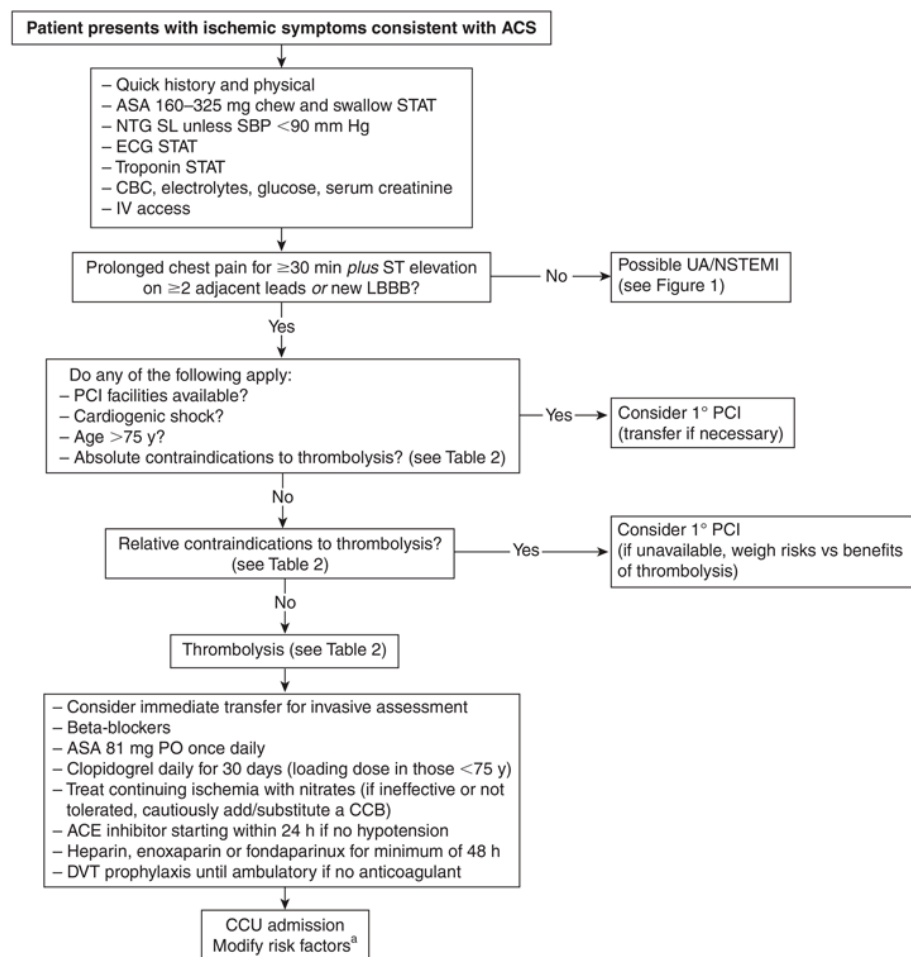


^[a] The recommended P2Y₁₂ inhibitors are clopidogrel and ticagrelor.

^[b] See [Post-myocardial Infarction](#).

Abbreviations: ACE = angiotensin-converting enzyme; ACS = acute coronary syndrome; ASA = acetylsalicylic acid; CABG = coronary artery bypass graft; CBC = complete blood count; CCB = calcium channel blocker; ECG = electrocardiogram; HF = heart failure; LBBB = left bundle branch block; LMWH = low-molecular-weight heparin; NTG = nitroglycerin; PCI = percutaneous coronary intervention; SBP = systolic blood pressure; STEMI = ST segment elevation myocardial infarction; TIMI = Thrombolysis in Myocardial Infarction; UA/NSTEMI = unstable angina or non-ST segment elevation myocardial infarction; UFH = unfractionated heparin; VT = ventricular tachycardia

Figure 2: Early Management of STEMI



^[a] See [Post-myocardial Infarction](#).

Abbreviations: ACE = angiotensin-converting enzyme; ACS = acute coronary syndrome; ASA = acetylsalicylic acid; CBC = complete blood count; CCB = calcium channel blocker; CCU = cardiac care unit; DVT = deep vein thrombosis; ECG = electrocardiogram; LBBB = left bundle branch block; NTG = nitroglycerin; PCI = percutaneous coronary intervention; SBP = systolic blood pressure; STEMI = ST segment elevation myocardial infarction; UA/NSTEMI = unstable angina or non-ST segment elevation myocardial infarction

Drug Table

Table 2: Drugs Used in Acute Coronary Syndromes

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Angiotensin-Converting Enzyme (ACE) Inhibitors				
captopril  generics	Initial: 6.25 mg Q8H PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
\$25–50	Target: 50 mg Q8H PO		Avoid concurrent therapy with lithium.	
cilazapril  Inhibace, generics	Initial: 0.5 mg daily PO Target: 2.5 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
< \$25			Avoid concurrent therapy with lithium.	

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
enalapril  Vasotec, generics < \$25	Initial: 2.5 mg BID PO Target: 10 mg BID PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
fosinopril generics < \$25	Initial: 5 mg daily PO Target: 20 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
lisinopril  Zestril, Prinivil, generics < \$25	Initial: 2.5 mg daily PO Target: 20–40 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness,	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride,	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or

		hyperkalemia, dry cough (common).	spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	single kidney with renal artery stenosis.
perindopril  Coversyl < \$25	Initial: 2 mg daily PO Target: 8 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
quinapril  Accupril, generics < \$25	Initial: 5 mg daily PO Target: 20–40 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
ramipril  Altace, generics < \$25	Initial: 1.25–2.5 mg BID PO Target: 5 mg BID or 10 mg daily PO	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.
trandolapril  Mavik, generics < \$25	Initial: 1 mg daily PO Target: 4 mg daily	Proteinuria (1%), neutropenia (rare), rash, angioedema, hypotension, alterations in taste, nausea, anorexia, dizziness, hyperkalemia, dry cough (common).	Increased risk of neutropenia with allopurinol, antiarrhythmics, corticosteroids. Hyperkalemia with amiloride, spironolactone, triamterene. Hypotension with diuretics. Avoid concurrent therapy with lithium.	Caution in patients with renovascular hypertension, renal insufficiency, bilateral renal artery stenosis or single kidney with renal artery stenosis.

Drug Class: Antiplatelet Agents

ASA Aspirin, generics < \$25	80–81 mg/day PO	Gastritis, gastric/duodenal ulceration (rarely bronchospasm). Nausea, vomiting, GI hemorrhage,	Bleeding increased with heparin (low risk) and warfarin; other NSAIDs .	
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		tinnitus, vertigo, hypersensitivity.		
clopidogrel Plavix, Clopidogrel, other generics < \$25	Loading dose: pre- PCI or STEMI : 300–600 mg PO UA/NSTEMI : 300 mg PO (omit if risk of bleeding high) Maintenance: 75 mg daily PO	Bleeding, rash, purpura. Similar tolerability to ASA .	Caution with NSAIDs . PPI use may reduce clopidogrel efficacy.	
prasugrel generics \$50–75	Loading dose: 60 mg, then 10 mg daily PO	Increased risk of bleeding.	Caution with NSAIDs .	Caution in patients >75 y, body weight <60 kg. Contraindicated if history of ischemic stroke.
ticagrelor Brilinta \$75–100	Loading dose: 180 mg, then 90 mg BID PO	Increased risk of bleeding, bradycardia, headache, nausea, transient dyspnea. Increased serum creatinine, uric acid.	Caution with NSAIDs . Avoid strong CYP3A4 inhibitors, e.g., clarithromycin. May increase digoxin levels.	

Drug Class: Beta₁-adrenergic Antagonists, selective with intrinsic sympathomimetic activity (ISA)

acebutolol  generics < \$25	Initial: 100–200 mg BID PO Target: 400 mg BID PO	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia,	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics.	
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Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		claudication, Raynaud phenomenon, lethargy, drowsiness.	Increased bradycardia with digoxin. Hypertension with alpha-agonists.	

Drug Class: Beta₁-adrenergic Antagonists, selective without ISA

atenolol  Tenormin, generics < \$25	Initial: 50 mg/day PO Target: 100 mg/day PO	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication, Raynaud phenomenon, lethargy, drowsiness.	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased bradycardia with digoxin. Hypertension with alpha-agonists.	
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bisoprolol  generics < \$25	Initial: 2.5 mg daily PO Target: 10 mg daily PO	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication, Raynaud phenomenon, lethargy, drowsiness.	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased bradycardia with digoxin. Hypertension with alpha-agonists.	
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metoprolol Lopresor, generics < \$25	IV: 5 mg over 1–2 min, repeat Q5 min (maximum 15 mg) Initial: 25–50 mg QID PO Target: 100 mg BID PO; PO can be started	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication,	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased	
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Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	15 min after IV	Raynaud phenomenon, lethargy, drowsiness.	bradycardia with digoxin. Hypertension with alpha-agonists.	

Drug Class: Beta₁-adrenergic Antagonists, nonselective without ISA

nadolol  generics < \$25	Initial: 40–80 mg/day PO Target: 160 mg/day PO	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication, Raynaud phenomenon, lethargy, drowsiness.	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased bradycardia with digoxin. Hypertension with alpha-agonists.	
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propranolol Inderal-LA, generics < \$25	Initial: 40 mg BID-TID PO Target: 60 mg QID PO or 240 mg daily PO as SR formulation	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication, Raynaud phenomenon, lethargy, drowsiness.	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased bradycardia with digoxin. Hypertension with alpha-agonists.	
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timolol generics < \$25	Initial: 5–10 mg BID PO Target: 10 mg BID PO	Bronchospasm, HF, hypotension, sleep disturbance, dizziness, fatigue, anorexia, nausea, AV block, bradycardia, claudication, Raynaud	Enhanced cardiodepressant effect with calcium channel blockers, antiarrhythmics, anesthetics. Increased bradycardia with	
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Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		phenomenon, lethargy, drowsiness.	digoxin. Hypertension with alpha-agonists.	
Drug Class: Calcium Channel Blockers				
amlodipine Norvasc, generics	5–10 mg/day PO	Hypotension, flushing, marked peripheral edema.		
< \$25				
diltiazem Tiazac, Tiazac XC, generics	120–360 mg/day PO Give IR formulations TID or QID; CD and XC once daily	Bradycardia, heart block, edema, hypotension.	Additive effect with beta-blockers, digoxin, amiodarone. Monitor for excessive bradycardia. May inhibit metabolism through CYP3A4	
< \$25				
verapamil Isoptin SR, generics	180–480 mg/day PO Give IR formulations TID; SR BID	Bradycardia, heart block, hypotension, constipation, flushing, edema.	Additive effect with beta-blockers, digoxin, amiodarone. Monitor for excessive bradycardia. May inhibit metabolism through CYP3A4	
< \$25				
Drug Class: Fibrinolytics				
alteplase Activase rt-PA	Bolus: 15 mg IV, then 0.75 mg/kg over 30 min (maximum 50 mg), then 0.5 mg/kg	Bleeding (can be fatal).		Absolute contraindications: pericarditis, previous intracranial
\$3,030/100 mg vial				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	over 60 min (maximum 35 mg)			hemorrhage; known malignant intracranial
	Maximum dose: 100 mg			neoplasm, known cerebral vascular lesion, ischemic stroke
	Start heparin with infusion			within 3 months except acute stroke within 3 h; suspected aortic dissection; active bleeding or bleeding diathesis (excluding menses); significant closed head or facial trauma within 3 months. Relative contraindications: history of chronic severe, poorly controlled HTN , severe uncontrolled HTN (BP >180/110 mm Hg) ^[b] ; previous CVA more than 3 months prior or known intracerebral pathology not covered above; traumatic or prolonged (>10 min) CPR or major surgery (<3 wk); noncompressible

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
				venous punctures; recent (2–4 wk) internal bleeding; pregnancy; active peptic ulcer; current use of anticoagulants.
tenecteplase TNKase \$3,030/(50 mg vial)	<60 kg: 30 mg IV × 1 60–69 kg: 35 mg IV × 1 70–79 kg: 40 mg IV × 1 80–89 kg: 45 mg IV × 1 ≥90 kg: 50 mg IV × 1 Given as IV bolus over 5 s	Bleeding (can be fatal).		Absolute contraindications: pericarditis, previous intracranial hemorrhage; known malignant intracranial neoplasm, known cerebral vascular lesion, ischemic stroke within 3 months except acute stroke within 3 h; suspected aortic dissection; active bleeding or bleeding diathesis (excluding menses); significant closed head or facial trauma within 3 months. Relative contraindications: history of chronic severe, poorly controlled HTN , severe uncontrolled HTN

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
				(BP >180/110 mm Hg) ^[b] ; previous CVA more than 3 months prior or known intracerebral pathology not covered above; traumatic or prolonged (>10 min) CPR or major surgery (<3 wk); noncompressible venous punctures; recent (2–4 wk) internal bleeding; pregnancy; active peptic ulcer; current use of anticoagulants.
Drug Class: Low Molecular Weight Heparins				
dalteparin  Fragmin < \$25 ^[c]	120 units/kg BID SC Maximum: 10 000 units/dose	Hematoma at injection site, bleeding, thrombocytopenia.	Bleeding increased with ASA (low risk) and warfarin.	Caution if ClCr <30 mL/min.
enoxaparin  Lovenox \$25–50 ^[c]	1 mg/kg BID SC Maximum: 100 mg for the first 2 doses Post-thrombolytic: 30 mg IV bolus in addition to first SC dose If ≥75 y, omit bolus and reduce dose	Hematoma at injection site, bleeding, thrombocytopenia.	Bleeding increased with ASA (low risk) and warfarin.	Caution in elderly patients and those with renal insufficiency. Preferred LMWH in ACS.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	to 0.75 mg/kg BID (75 mg maximum for first 2 doses)			
Drug Class: Mineralocorticoid Receptor Antagonists				
eplerenone  Inspra, generics \$50–75	Initial: 25 mg daily PO Target: 50 mg daily PO	Hyperkalemia, dehydration, dizziness, diarrhea, nausea.	Hyperkalemia with ACE inhibitors, ARBs , NSAIDs . Strong inhibitors/inducers of CYP3A4.	
spironolactone  Aldactone, generics < \$25	Initial: 12.5 mg daily PO Target: 25–50 mg daily PO	Rash, urticaria, gynecomastia, nausea, vomiting, diarrhea, confusion, hyperkalemia, agranulocytosis, SLE .	Hyperkalemia with ACE inhibitors, ARBs , digoxin, NSAIDs .	
Drug Class: Nitrates				
isosorbide dinitrate generics < \$25	Initial: 30–90 mg/day PO Divide dose TID (allow a 12-h nitrate-free period)	Headache (up to 50%; tolerance may develop), tachycardia, palpitation, hypotension, syncope (rare), dizziness, nausea, flushing, weakness.	Potential hypotensive effect with vasodilators. Contraindicated with recent (<24 h) use of sildenafil, tadalafil or vardenafil.	Use with caution when hypotension present.
isosorbide-5- mononitrate Imdur, other generics < \$25	30–60 mg daily PO	Headache (up to 50%; tolerance may develop), tachycardia, palpitation, hypotension,	Potential hypotensive effect with vasodilators. Contraindicated with recent (<24 h) use of	Use with caution when hypotension present.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		syncope (rare), dizziness, nausea, flushing, weakness.	sildenafil, tadalafil or vardenafil.	
nitroglycerin, intravenous generics < \$25 ^[a]	Initial: 10– 150 mcg/min IV (titrate to symptoms and blood pressure)	Headache (up to 50%; tolerance may develop), tachycardia, palpitation, hypotension, syncope (rare), dizziness, nausea, flushing, weakness.	Potential hypotensive effect with vasodilators. Contraindicated with recent (<24 h) use of sildenafil, tadalafil or vardenafil.	Frequent BP monitoring is required for IV nitroglycerin.
nitroglycerin, sublingual Nitrostat, Nitrolingual Pumpspray, generics < \$25/container	Tablet: 0.3–0.6 mg SL PRN Q5 min Spray: 0.4 mg SL PRN Q5 min	Headache (up to 50%; tolerance may develop), tachycardia, palpitation, hypotension, syncope (rare), dizziness, nausea, flushing, weakness.	Potential hypotensive effect with vasodilators. Contraindicated with recent (<24 h) use of sildenafil, tadalafil or vardenafil.	Use with caution when hypotension present.
nitroglycerin, transdermal Minitran, Nitro- Dur, Transderm- Nitro, Trinipatch, generics \$25–50	0.2–0.8 mg/h patch applied daily for 10– 12 h topically	Headache (up to 50%; tolerance may develop), tachycardia, palpitation, hypotension, syncope (rare), dizziness, nausea, flushing, weakness. Contact dermatitis.	Potential hypotensive effect with vasodilators. Contraindicated with recent (<24 h) use of sildenafil, tadalafil or vardenafil.	Use with caution when hypotension present.
Drug Class: Specific Factor Xa Inhibitors				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
fondaparinux  Arixtra, generics < \$25 ^[c]	2.5 mg daily SC First dose IV if STEMI	Bleeding, allergic reactions (rare).	Use caution with other drugs that affect hemostasis.	Not recommended as sole agent in patients undergoing PCI due to risk of catheter thrombosis.

Drug Class: Unfractionated Heparin

heparin, unfractionated Heparin Sodium Injection, generics < \$25 /20 000 unit vial	60–70 units/kg IV bolus Maximum: 5000 units, then 12–15 units/kg/h Adjust to maintain aPTT at 1.5–2 × control First aPTT 4 h after bolus Post-thrombolytic: bolus 60 units/kg IV Maximum: 4000 units, then 12 units/kg/h Adjust to maintain aPTT at 1.5–2 × control First aPTT 6 h after bolus	Bleeding, thrombocytopenia.	Bleeding increased with ASA (low risk) and warfarin; aPTT response may be blunted with concurrent IV nitroglycerin (controversial).	
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^[a] Cost of 30-day supply unless otherwise specified; includes drug cost only.

^[b] Could be an absolute contraindication in low-risk patients with STEMI.

^[c] Cost of average 1-day supply.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).
Abbreviations: ACE = angiotensin-converting enzyme; ACS = acute coronary syndromes; aPTT = activated partial thromboplastin time; ARB = angiotensin receptor blocker; AV = atrioventricular; BP = blood pressure; CPR = cardiopulmonary resuscitation; CVA = cerebrovascular accident; GI = gastrointestinal; HF = heart

failure; HTN = hypertension; LMWH = low-molecular-weight heparin; NSAID = nonsteroidal anti-inflammatory drug; PCI = percutaneous coronary intervention; PPI = proton pump inhibitor; SLE = systemic lupus erythematosus; STEMI = ST segment elevation myocardial infarction; UA/NSTEMI = unstable angina or non-ST segment elevation myocardial infarction

Suggested Readings

[Mehta SR, Bainey KR, Cantor WJ et al. 2018 Canadian Cardiovascular Society/Canadian Association of Interventional Cardiology focused update of the guidelines for the use of antiplatelet therapy. *Can J Cardiol* 2018;34\(3\):214–33.](#)

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